

■ React Mastery Roadmap

This document summarizes the complete React learning journey, covering all major concepts from basics to advanced topics, with theory, examples, and best practices.

1. State Management

- useState, useReducer
- State vs Props
- Lifting State Up
- Patterns for Global State

2. React Hooks

- useState, useEffect, useContext, useMemo, useCallback, useRef
- Custom Hooks & Hook Rules

3. Event Handling

- Synthetic Events
- Event Binding
- Event Parameters

4. Conditional Rendering

- if/else, Ternary, Logical &&, Switch

5. Lists & Keys

- Rendering Lists
- Keys & Performance
- Dynamic Lists

6. Forms

- Controlled vs Uncontrolled Components
- Form Validation
- Formik & React Hook Form

7. Context API

- createContext, Provider, Consumer, useContext Hook
- Best Practices

8. State Management Libraries

- Redux & Redux Toolkit

- Zustand
- Recoil
- Context + useReducer

9. React Router

- Installation & Setup
- Nested Routes
- Navigation (Link, NavLink, useNavigate)
- Route Guards & Dynamic Routing

10. HTTP Requests & APIs

- Fetch API
- Axios
- Loading/Error Handling
- Custom API Hooks

11. Performance Optimization

- React.memo, useMemo, useCallback
- React.lazy, Suspense
- Code Splitting, Profiling

12. Error Boundaries

- Error Boundary Class
- componentDidCatch, getDerivedStateFromError
- Libraries for Error Boundaries

13. Refs & DOM Access

- useRef Hook
- Forwarding Refs
- Callback Refs
- DOM Manipulation

14. Testing

- Jest, React Testing Library
- Component & Hook Testing
- Integration, Snapshot, E2E (Cypress, Playwright)

15. Build Tools & Bundlers

- Webpack, Vite, Parcel, Rollup, ESBuild

16. Development Tools

- React DevTools
- ESLint, Prettier
- Husky, Storybook
- Strict Mode

17. Deployment

- Build Process
- Netlify, Vercel, GitHub Pages
- CDN Deployment
- CI/CD Pipelines

18. Architecture Patterns

- Component Architecture
- Folder Structure
- State Management Strategy
- Design Patterns

19. Security

- XSS Prevention, CSRF Protection
- Input Sanitization
- Secure Coding Practices

■ Congratulations! You have completed your React Mastery Journey and are now confident to build production-grade applications.